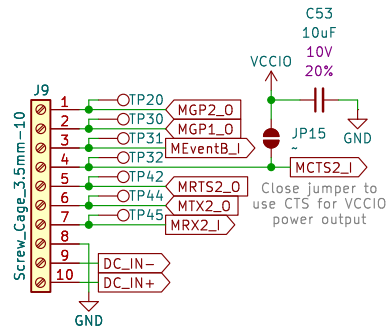
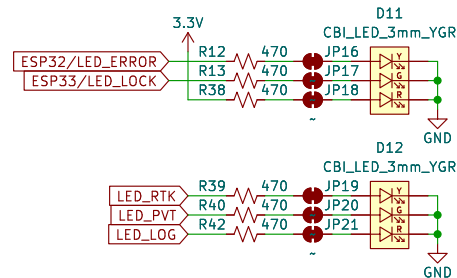


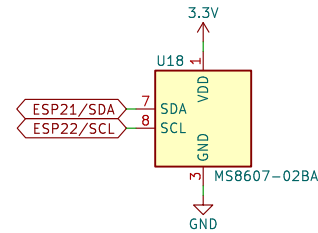
I/O Connector



LEDs

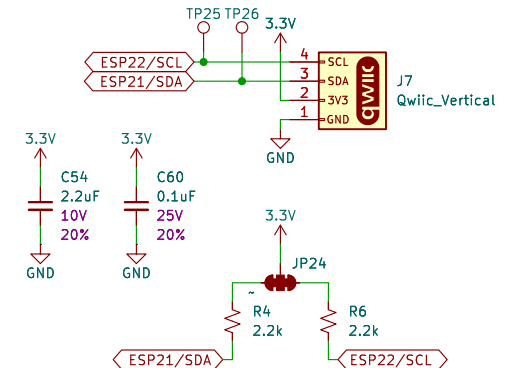


PHT Sensor – MS8607-02BA



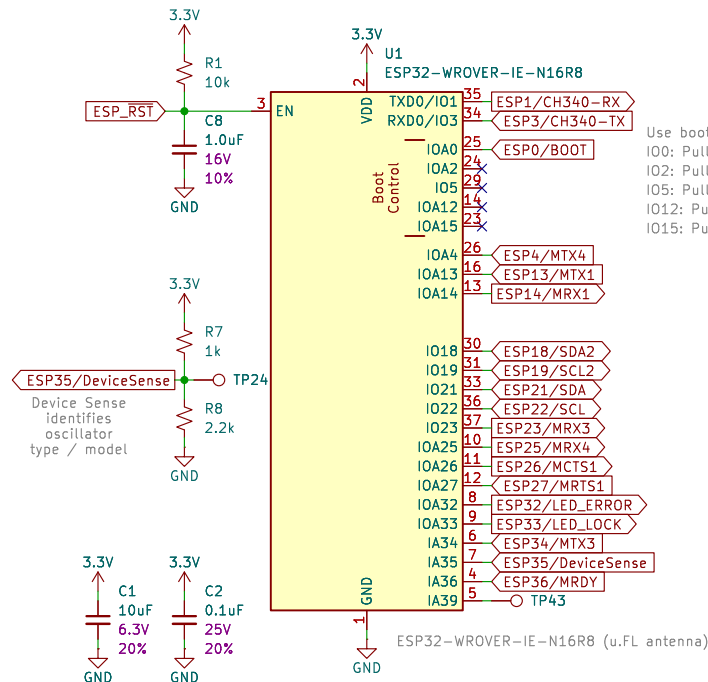
MS8607 I²C addresses:
P&T: 0x76 (unshifted)
H: 0x40 (unshifted)

Qwiic I²C (for OLED)



SSD1306 default I²C address: 0x3D (unshifted)

ESP32-WROVER



Power	PowerPath
File: Power.kicad_sch	File: PowerPath.kicad_sch
USB	
File: USB.kicad_sch	
GNSS	
File: GNSS.kicad_sch	
Ethernet	
File: Ethernet.kicad_sch	
LevelShifting	
File: LevelShifting.kicad_sch	
LevelShifting_10MHz	
File: LevelShifting_10MHz.kicad_sch	
Oscillator	
File: Oscillator.kicad_sch	



SPARKPNT

Designed by: P.C.
SparkFun Electronics

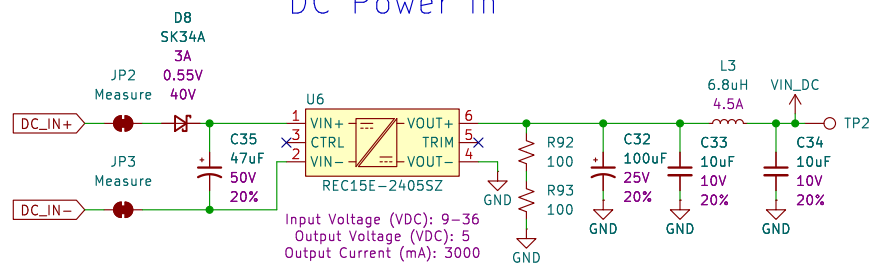
Sheet: /
File: SparkPNT_GNSSDO_Plus.kicad_sch

Title: GNSSDO Plus (mosaic-T, STP3593LF)

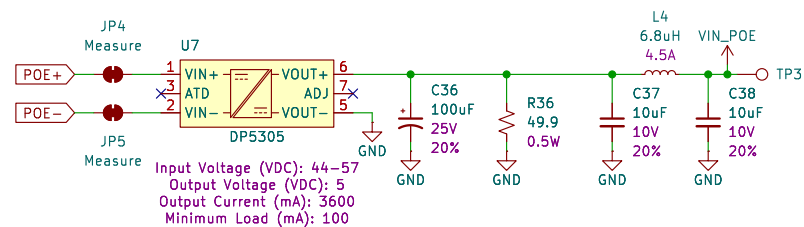
Size: USLetter Date: 2026-04-17
KiCad E.D.A. 9.0.8

Rev: v10
Id: 1/9

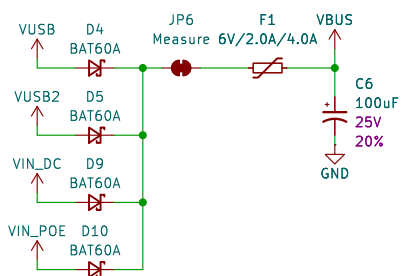
DC Power In



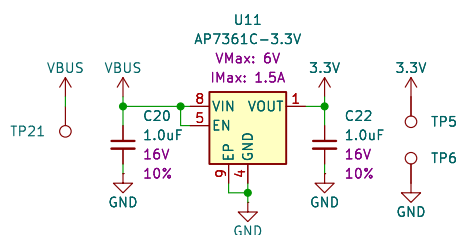
Power Over Ethernet



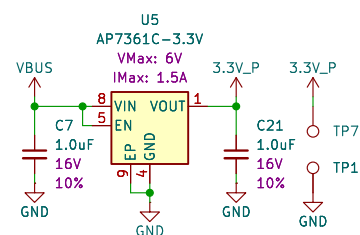
Power Mux



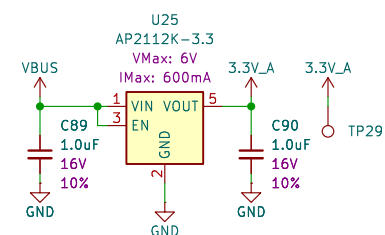
Main 3.3V



Peripheral 3.3V



Analog 3.3V



Sheet: /Power/
File: Power.kicad_sch

Title: GNSSDO Plus - Power

Size: USLetter Date:

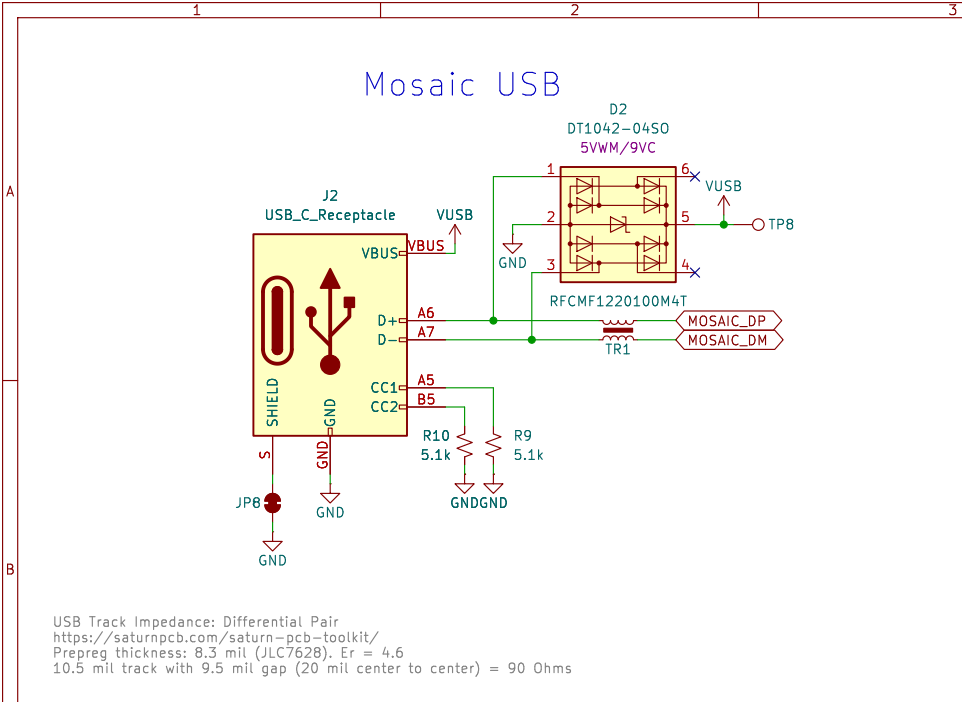
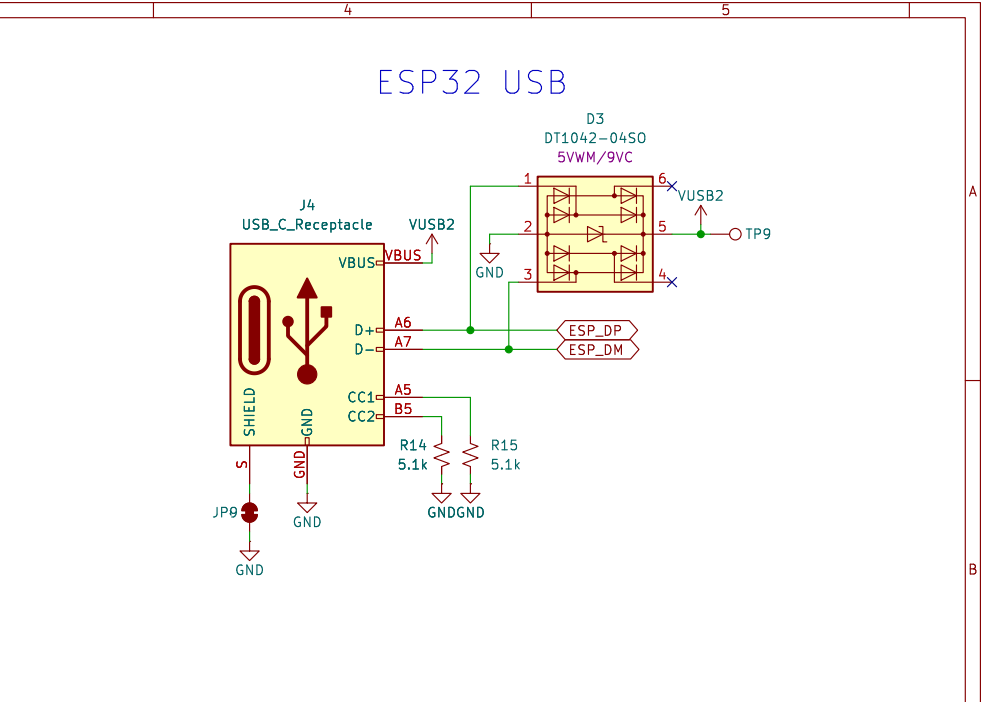
KiCad E.D.A. 9.0.8

Rev:

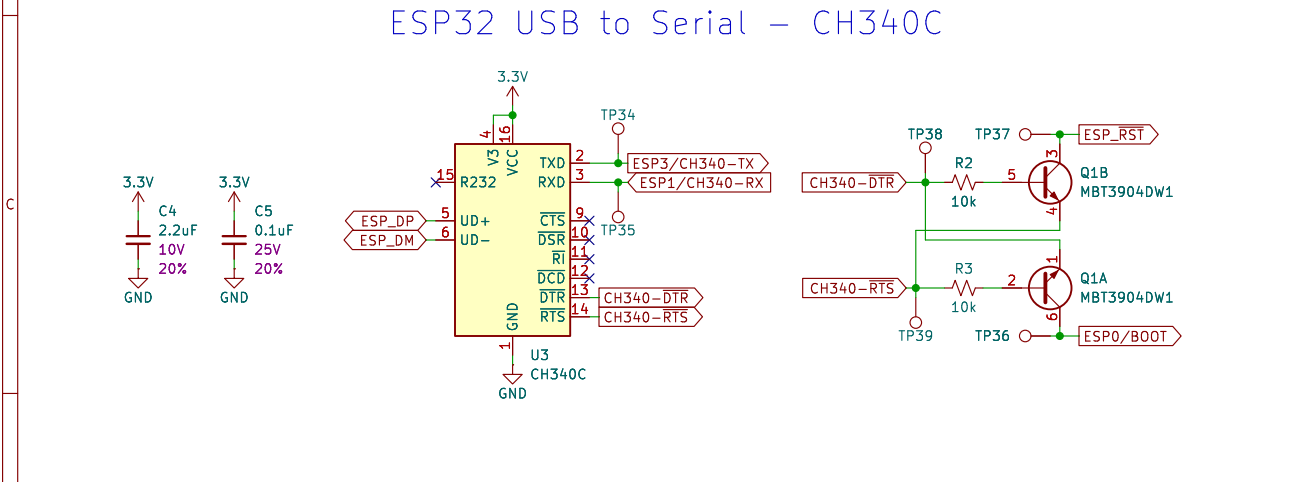
Id: 2/9

Mosaic USB

USB Track Impedance: Differential Pair
<https://saturnpcb.com/saturn-pcb-toolkit/>
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
10.5 mil track with 9.5 mil gap (20 mil center to center) = 90 Ohms

[illegible][illegible]

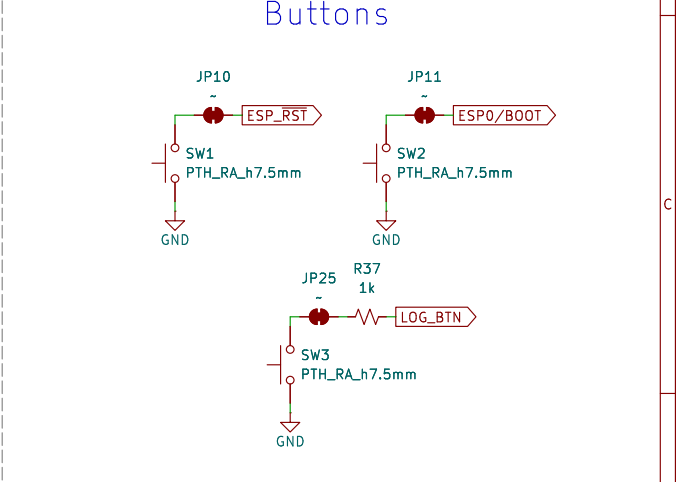
The diagram illustrates the internal circuitry of the ESP32 USB to Serial module (CH340C). It shows the power supply section with two 3.3V regulators (C4 and C5) connected to GND. The main CH340C chip (U3) is shown with its pins connected to the ESP32's pins. The ESP32 pins are labeled: ESP_DP (pin 5), ESP_DM (pin 6), ESP3/CH340-TX (pin 2), ESP1/CH340-RX (pin 3), CH340-DTR (pin 13), CH340-RTS (pin 14), and ESP0/BOOT (pin 6). The CH340C chip has pins: R232 (pin 15), VCC (pin 4), TXD (pin 2), RXD (pin 3), CTS (pin 9), DSR (pin 10), RI (pin 11), DCD (pin 12), DTR (pin 13), RTS (pin 14), and GND (pin 1). The signal routing section shows the CH340C's DTR and RTS pins connected to the ESP32's DTR and RTS pins via resistors R2 and R3 (10k). The CH340C's TXD and RXD pins are connected to the ESP32's TX and RX pins. The CH340C's VCC pin is connected to the 3.3V supply. The CH340C's GND pin is connected to GND. The CH340C's CTS and DSR pins are connected to the ESP32's CTS and DSR pins. The CH340C's RI and DCD pins are connected to the ESP32's RI and DCD pins. The CH340C's RST pin is connected to the ESP32's RST pin. The CH340C's BOOT pin is connected to the ESP32's BOOT pin. The CH340C's DTR and RTS pins are connected to the ESP32's DTR and RTS pins via resistors R2 and R3 (10k). The CH340C's TXD and RXD pins are connected to the ESP32's TX and RX pins. The CH340C's VCC pin is connected to the 3.3V supply. The CH340C's GND pin is connected to GND. The CH340C's CTS and DSR pins are connected to the ESP32's CTS and DSR pins. The CH340C's RI and DCD pins are connected to the ESP32's RI and DCD pins. The CH340C's RST pin is connected to the ESP32's RST pin. The CH340C's BOOT pin is connected to the ESP32's BOOT pin.



Buttons

The image shows three circuit diagrams for buttons, labeled JP10, JP11, and JP25.

- JP10:** A push button connected to the `ESP_RST` pin. The button is labeled `SW1` and `PTH_RA_h7.5mm`. It is connected to `GND` through a pull-up resistor `R37` (1k).
- JP11:** A push button connected to the `ESP0/BOOT` pin. The button is labeled `SW2` and `PTH_RA_h7.5mm`. It is connected to `GND` through a pull-up resistor `R37` (1k).
- JP25:** A momentary button connected to the `LOG_BTN` pin. The button is labeled `SW3` and `PTH_RA_h7.5mm`. It is connected to `GND` through a pull-up resistor `R37` (1k).



The schematic diagram illustrates the internal components and connections of the mosaic Tri-band GNSS module. The central component is the MOSAIC-T module (U2), which is connected to various external components and interfaces.

Power and Ground Connections:

- 3.3V_P:** Connected to VDD_3V3, VDD_BAT, and VUSB.
- 2.5V:** Connected to C3 (0.1uF) and R11 (10k).
- 1.8V:** Connected to C18 (2.2uF) and C19 (0.1uF).
- 2.8V:** Connected to C73 (2.2uF) and C74 (0.1uF).
- 6.3V:** Connected to C29 (22uF).
- 10V:** Connected to C16 (2.2uF) and C17 (0.1uF).

Signal and Data Connections:

- ETH_CLK (50MHz):** Connected to ETH_CLK.
- ETH_MDIO:** Connected to ETH_MDIO.
- ETH_MDC:** Connected to ETH_MDC.
- ETH_RXD0, ETH_RXD1, ETH_RXD2:** Connected to ETH_RXD0, ETH_RXD1, and ETH_RXD2.
- ETH_TXD0, ETH_TXD1, ETH_TXD2:** Connected to ETH_TXD0, ETH_TXD1, and ETH_TXD2.
- ETH_TXEN:** Connected to ETH_TXEN.
- ETH_RXDV:** Connected to ETH_RXDV.
- ETH_RST:** Connected to ETH_RST.
- 10MHz_Mosaic:** Connected to AC17 and AC18.
- TP22:** Connected to TP22.
- TP41:** Connected to TP41.
- TP23:** Connected to TP23.
- TP24:** Connected to TP24.
- TP25:** Connected to TP25.
- TP26:** Connected to TP26.
- TP27:** Connected to TP27.
- TP28:** Connected to TP28.
- TP29:** Connected to TP29.
- TP30:** Connected to TP30.
- TP31:** Connected to TP31.
- TP32:** Connected to TP32.
- TP33:** Connected to TP33.
- TP34:** Connected to TP34.
- TP35:** Connected to TP35.
- TP36:** Connected to TP36.
- TP37:** Connected to TP37.
- TP38:** Connected to TP38.
- TP39:** Connected to TP39.
- TP40:** Connected to TP40.
- TP41:** Connected to TP41.
- TP42:** Connected to TP42.
- TP43:** Connected to TP43.
- TP44:** Connected to TP44.
- TP45:** Connected to TP45.
- TP46:** Connected to TP46.
- TP47:** Connected to TP47.
- TP48:** Connected to TP48.
- TP49:** Connected to TP49.
- TP50:** Connected to TP50.
- TP51:** Connected to TP51.
- TP52:** Connected to TP52.
- TP53:** Connected to TP53.
- TP54:** Connected to TP54.
- TP55:** Connected to TP55.
- TP56:** Connected to TP56.
- TP57:** Connected to TP57.
- TP58:** Connected to TP58.
- TP59:** Connected to TP59.
- TP60:** Connected to TP60.
- TP61:** Connected to TP61.
- TP62:** Connected to TP62.
- TP63:** Connected to TP63.
- TP64:** Connected to TP64.
- TP65:** Connected to TP65.
- TP66:** Connected to TP66.
- TP67:** Connected to TP67.
- TP68:** Connected to TP68.
- TP69:** Connected to TP69.
- TP70:** Connected to TP70.
- TP71:** Connected to TP71.
- TP72:** Connected to TP72.
- TP73:** Connected to TP73.
- TP74:** Connected to TP74.
- TP75:** Connected to TP75.
- TP76:** Connected to TP76.
- TP77:** Connected to TP77.
- TP78:** Connected to TP78.
- TP79:** Connected to TP79.
- TP80:** Connected to TP80.
- TP81:** Connected to TP81.
- TP82:** Connected to TP82.
- TP83:** Connected to TP83.
- TP84:** Connected to TP84.
- TP85:** Connected to TP85.
- TP86:** Connected to TP86.
- TP87:** Connected to TP87.
- TP88:** Connected to TP88.
- TP89:** Connected to TP89.
- TP90:** Connected to TP90.
- TP91:** Connected to TP91.
- TP92:** Connected to TP92.
- TP93:** Connected to TP93.
- TP94:** Connected to TP94.
- TP95:** Connected to TP95.
- TP96:** Connected to TP96.
- TP97:** Connected to TP97.
- TP98:** Connected to TP98.
- TP99:** Connected to TP99.
- TP100:** Connected to TP100.

Module Interfaces:

- ESP36/MDY:** Connected to ESP36/MDY.
- ESP14/MRX1, ESP13/MTX1, ESP27/MRTS1, ESP26/MCTS1:** Connected to ESP14/MRX1, ESP13/MTX1, ESP27/MRTS1, and ESP26/MCTS1.
- MRX2, MTX2, MRTS2, MCTS2:** Connected to MRX2, MTX2, MRTS2, and MCTS2.
- ESP23/MRX3, ESP34/MTX3:** Connected to ESP23/MRX3 and ESP34/MTX3.
- ESP25/MRX4, ESP4/MTX4:** Connected to ESP25/MRX4 and ESP4/MTX4.
- SD_CLK, SD_CMD, SD_DATA:** Connected to SD_CLK, SD_CMD, and SD_DATA.
- LOG_BTN, LOG_LED:** Connected to LOG_BTN and LOG_LED.
- MPPS_1V8:** Connected to MPPS_1V8.
- MEventA_1V8, MEventB_1V8:** Connected to MEventA_1V8 and MEventB_1V8.
- MGP1, MGP2, LED_PVT, LED_RTK:** Connected to MGP1, MGP2, LED_PVT, and LED_RTK.

Other Connections:

- COM1_RX, COM1_TX, COM1_RTS, COM1_CTS:** Connected to COM1_RX, COM1_TX, COM1_RTS, and COM1_CTS.
- COM2_RX, COM2_TX, COM2_RTS, COM2_CTS:** Connected to COM2_RX, COM2_TX, COM2_RTS, and COM2_CTS.
- COM3_RX, COM3_TX, COM3_RTS, COM3_CTS:** Connected to COM3_RX, COM3_TX, COM3_RTS, and COM3_CTS.
- COM4_RX, COM4_TX:** Connected to COM4_RX and COM4_TX.
- RTC_XTALI, RTC_XTALO:** Connected to RTC_XTALI and RTC_XTALO.
- REF_IN, REF_OUT:** Connected to REF_IN and REF_OUT.
- 2V8_IN, 2V8_OUT:** Connected to 2V8_IN and 2V8_OUT.
- 1V8_OUT, SYNC:** Connected to 1V8_OUT and SYNC.
- ONOFF, PMIC_ON_REQ:** Connected to ONOFF and PMIC_ON_REQ.

[illegible]

GNSS Antenna

The diagram illustrates a GNSS antenna circuit. It features a microstrip line of 500hm length connected to a SMA connector (J1). The line is terminated at the MOSAIC_RF_IN port. A diode (D1, PESD0402, 14VWM/40VC) is connected in series to ground (GND).

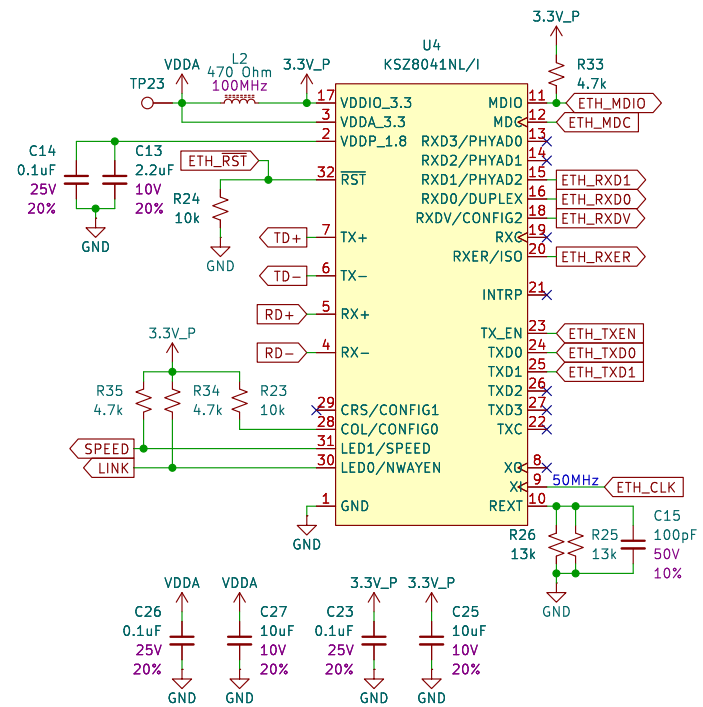
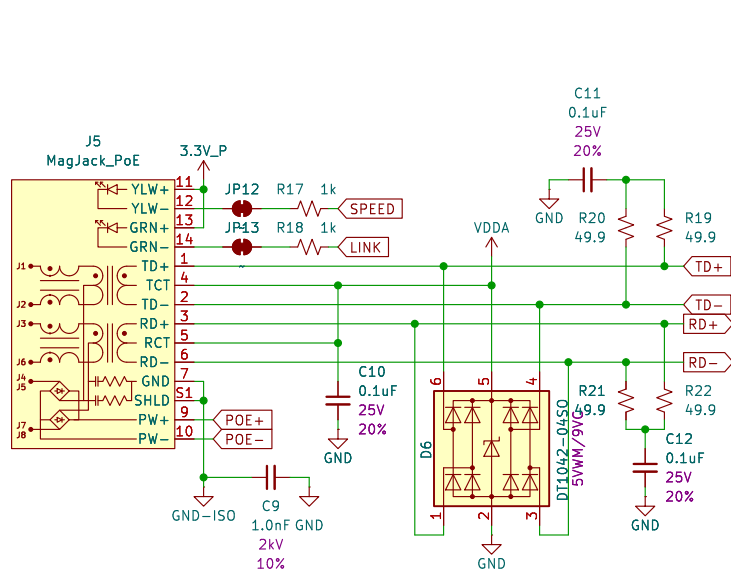
Microstrip Calculation:

- Copper Thickness (1oz): $1.4\text{mil}/0.035\text{mm}$
- Board thickness: 1.6mm
- Dielectric thickness (layer 1 to 2): 0.2mm
- Er: 4.6
- Polygon Isolation: $6\text{mil}/0.1524\text{mm}$
- RF Trace Width: $13\text{mil}/0.33\text{mm}$

<https://chemandy.com/calculators/coplanar-waveguide-with-ground-calculator.htm>

Rev:
Id: 4/9

Ethernet



Ethernet Track Impedance: Differential Pair
<https://saturnpcb.com/saturn-pcb-toolkit/>
 Prepreg thickness: 8.3 mil (JLC7628), Er = 4.6
 9.0 mil track with 11.0 mil gap (20 mil center to center) = 100 Ohms
 Each pair should match in length to better than 0.5mm

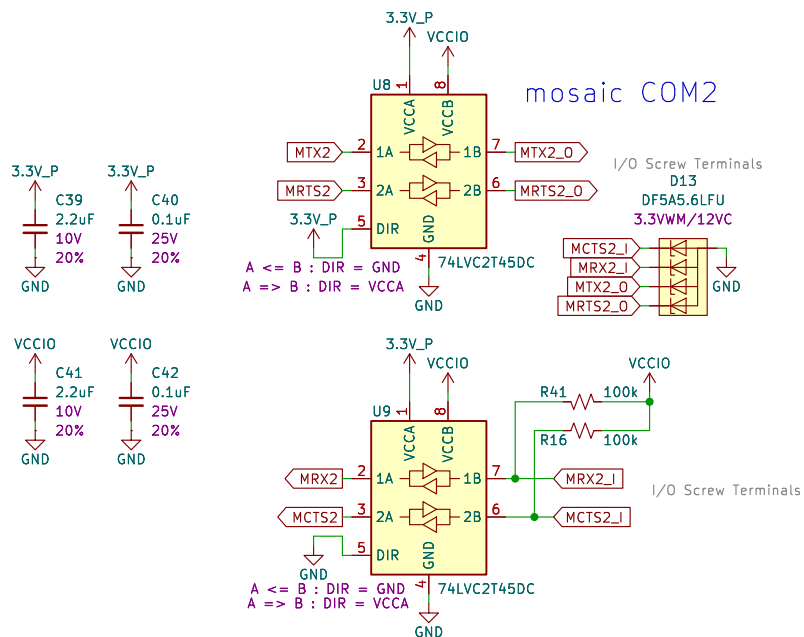
Sheet: /Ethernet/
 File: Ethernet.kicad_sch

Title: GNSSDO Plus - Ethernet

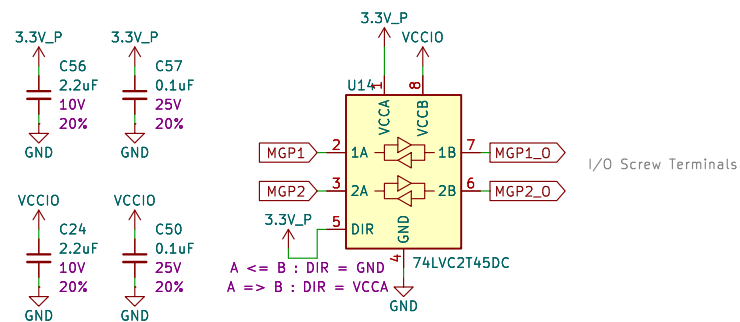
Size: USLetter Date:
 KiCad E.D.A. 9.0.8

Rev:
 Id: 5/9

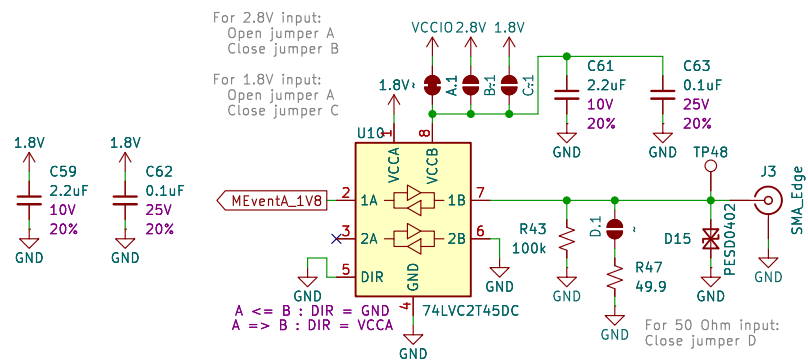
mosaic COM2



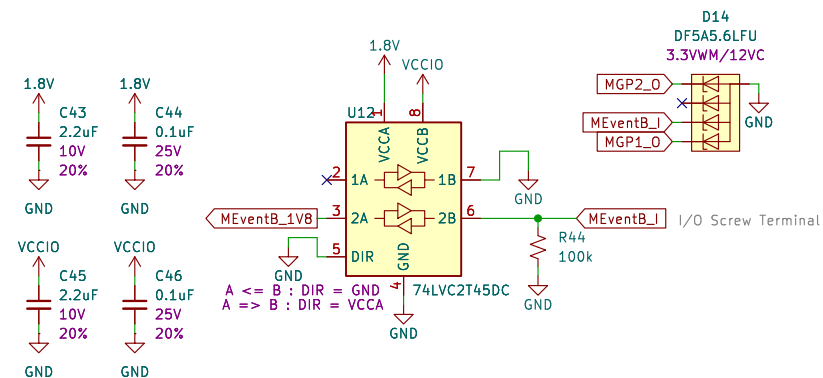
GP1/2 Out



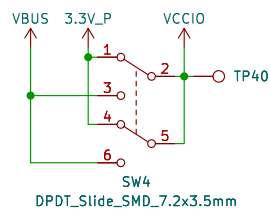
EventA In



EventB In



VCCIO voltage selection



Level-Shifting

Sheet: /LevelShifting/
File: LevelShifting.kicad_sch

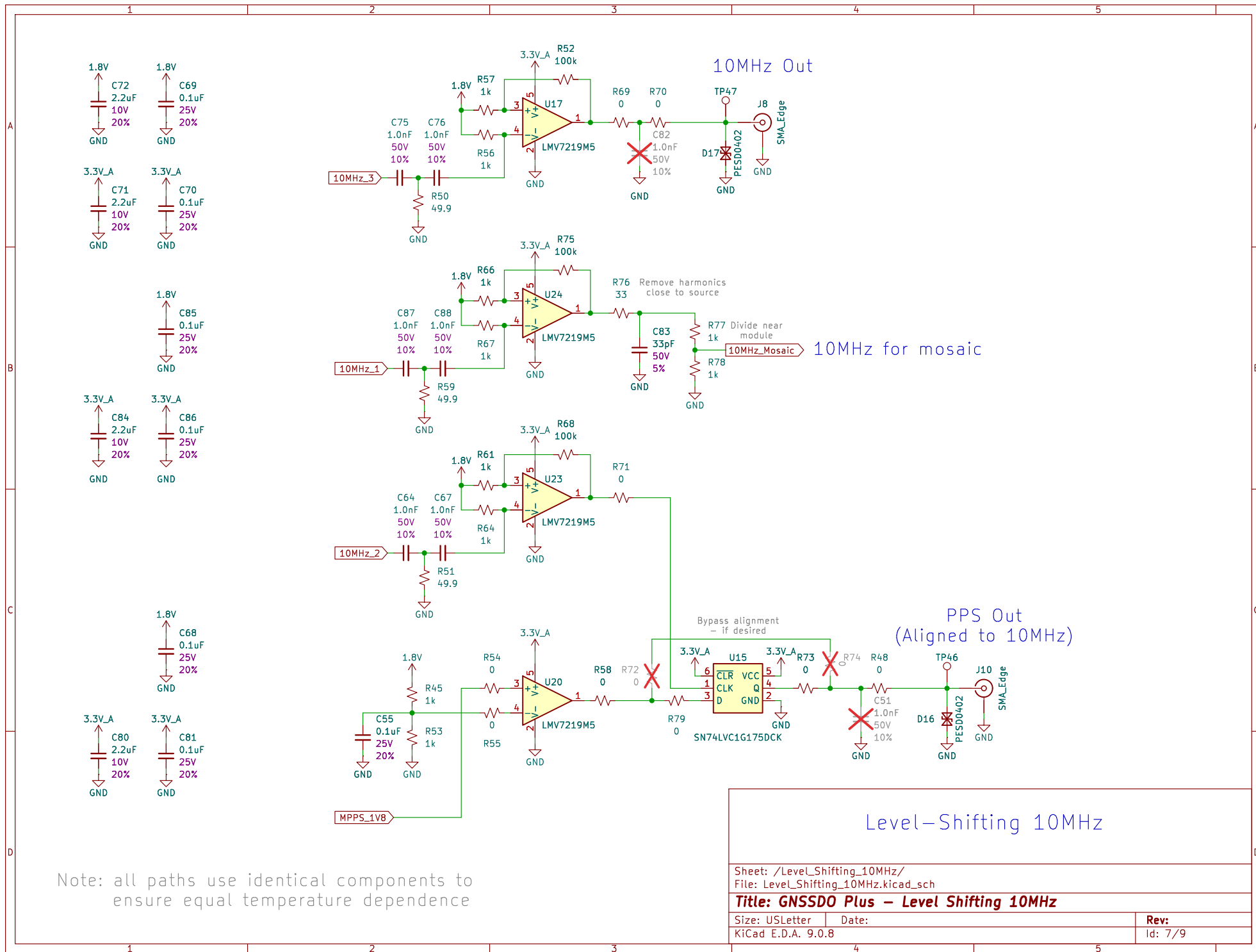
Title: GNSSDO Plus – Level Shifting

Size: USLetter Date:

KiCad E.D.A. 9.0.8

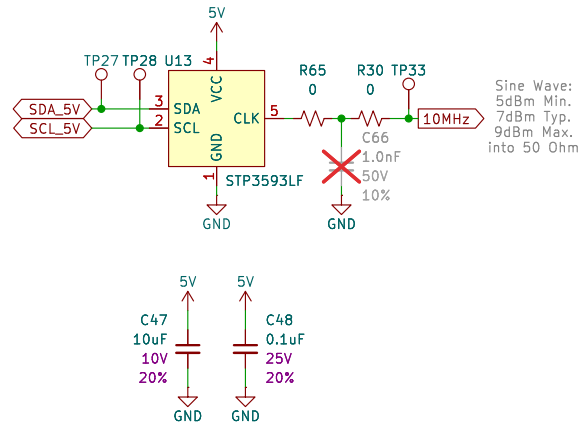
Rev:

Id: 6/9



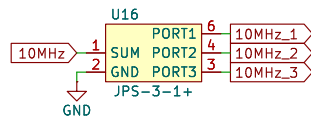
10MHz Oscillator – STP3593LF

Supply Voltage: 5.0V (4.75V Min., 5.25V Max.)
Current Consumption: 1500mA (Warm Up), 600mA (Steady State)

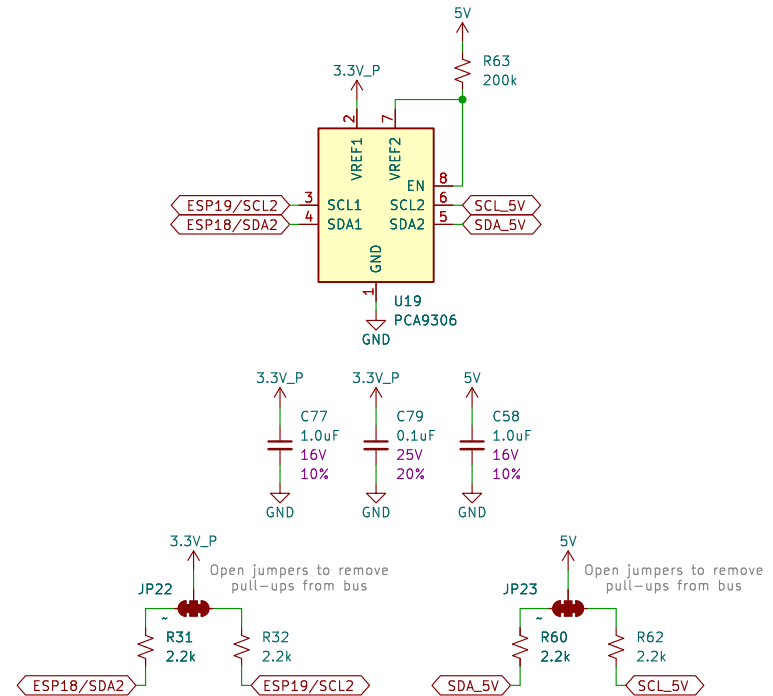


3-Way Splitter – JPS-3-1+

Typical Total Loss: 5.0dB at 10MHz



I²C Level Shifting – PCA9306



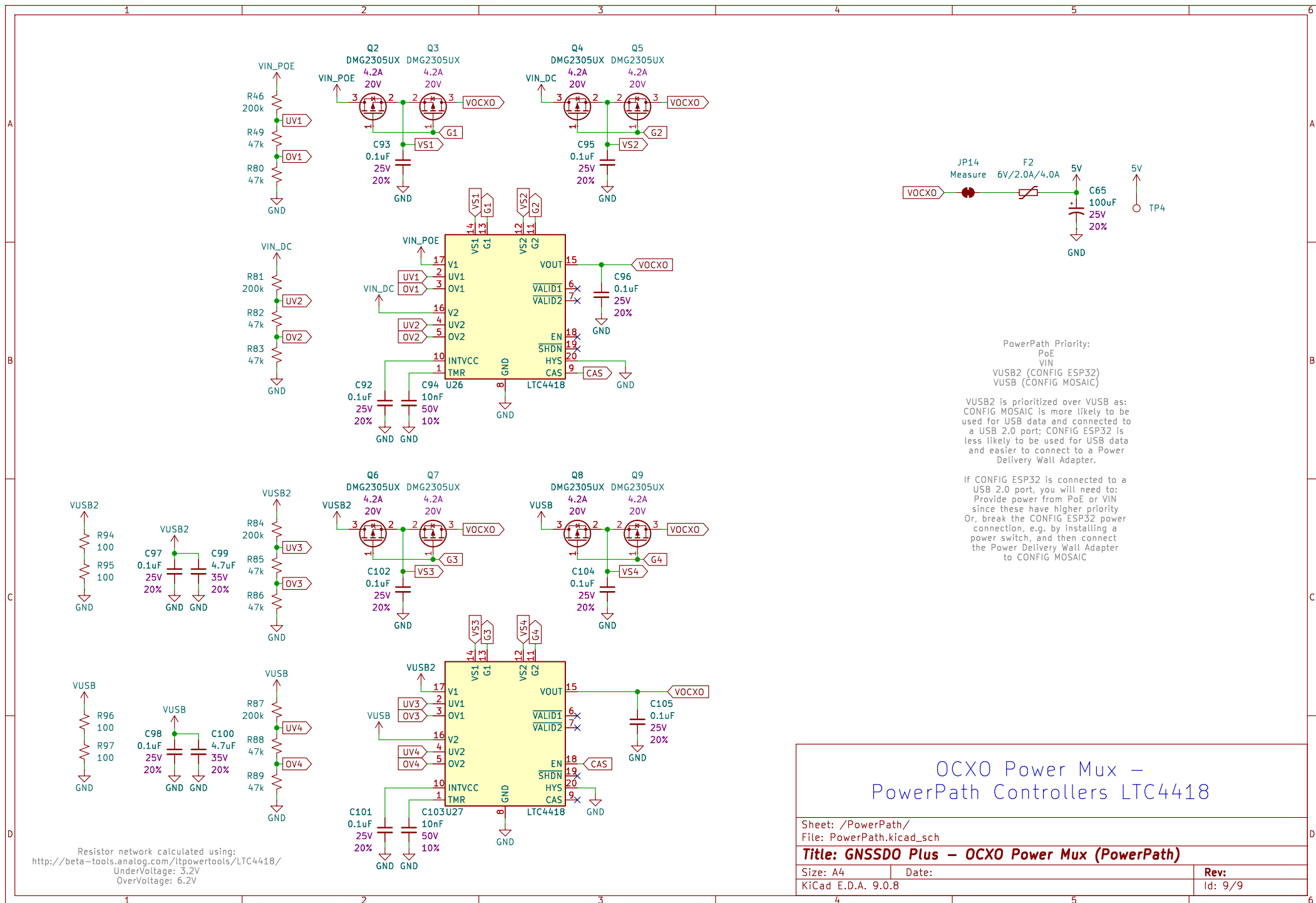
Sheet: /Oscillator/
File: Oscillator.kicad_sch

Title: GNSSDO Plus – Oscillator

Size: A4
KiCad E.D.A. 9.0.8

Date:

Rev:
Id: 8/9



OCXO Power Mux – PowerPath Controllers LTC4418

Sheet: /PowerPath/
 File: PowerPath.kicad_sch

Title: GNSSDO Plus – OCXO Power Mux (PowerPath)

Size: A4
 KiCad E.D.A. 9.0.8

Date:

Rev:
 Id: 9/9